

RAJA SANA ULLAH

Electrical Engineer

☎ +92-3344477194 ✉ sanaullahraja05@gmail.com 🔗 linkedin.com/in/raja-sana-ullah

SUMMARY

Electrical Engineering graduate from the University of Engineering and Technology (UET), Lahore, with hands-on exposure to power systems, switchgear, and transformer engineering gained through internships at Pak Elektron Limited (PEL) and IESCO 132 kV Grid Station. Skilled in power system load flow analysis, fault studies, and protection coordination using MATLAB/Simulink and ETAP, with a strong academic foundation in Power Electronics, Control Systems, and Smart Grids. Published researcher in DC-DC converter control and stability analysis, with additional experience in digital IC design (Verilog) and machine learning. Seeking to apply practical power systems and switchgear knowledge in an industry engineering role focused on power distribution and protection systems.

PROFESSIONAL EXPERIENCE

Pak Elektron Limited

Internee as a Design Engineer (Electrical), Transformer Design Department Aug 2024 – Sep 2024

- **Transformer Fundamentals & Applications:** Gained foundational knowledge on transformer types, uses and manufacturing process, building a comprehensive understanding of their role in power systems.
- **Transformer Design Principals:** Acquired practical skills in transformer design including key design parameters and methods used to meet specific distribution power requirements.

IESCO 132kV Grid Station, Jhelum

Electrical Engineering Intern June 2025 – July 2025

- **Grid Operations:** Assisted in monitoring and operation of 132kV equipment including transformers, circuit breakers, isolators, and protection panels.
- **Maintenance Support:** Participated in inspections and preventive maintenance to ensure reliable power supply.
- **Protection Systems:** Gained exposure to relay coordination, protection schemes, and control room operations.
- **Fault Handling:** Supported fault diagnosis, troubleshooting, and restoration activities.
- **Safety Practices:** Followed electrical safety standards and grid station operational procedures.

Elevvo Pathway (Egypt) — Remote Internship

Machine Learning Engineer Intern July 2025 – Aug 2025

- **Model Development:** Built and trained ML models using Python (NumPy, Pandas, Scikit-learn) for prediction and classification tasks.
- **Data Processing:** Performed data cleaning, feature engineering, and EDA to improve model performance.
- **Evaluation & Tuning:** Applied performance metrics and hyperparameter tuning for optimization.
- **Documentation:** Prepared technical reports and assisted in basic deployment concepts.
- **Remote Collaboration:** Participated in meetings, code reviews, and team discussions in a remote work environment.

EDUCATION

Bachelor of Electrical Engineering

University of Engineering & Technology Lahore, Pakistan
Lahore, Pakistan

Dec 2022 – June 2026

- CGPA: 3.2/4.00
- Course Work: Power Electronics, Control Systems, Power System Analysis & Design, Power System Operation & Control, Smart Grid
- Thesis: Piezo Electric Based Smart Helmet

ACADEMIC EXPERIENCE

Piezoelectric-Based Smart Helmet (Final Year Project)

Oct 2025 – May 2026

I designed this project during my final year; it is a Piezoelectric-Based Smart Helmet that enhances worker safety by detecting impacts and vibrations. The helmet provides automatic alerts, making it a practical and innovative safety solution. Here is the summary of this project:

- **Energy Harvesting Design:** Developed a smart helmet using piezoelectric sensors to generate electrical energy from vibrations and mechanical movements.
- **Safety & Monitoring:** Integrated sensing mechanisms to enhance worker safety and enable real-time monitoring of impact and abnormal conditions.
- **Embedded Implementation:** Designed microcontroller-based circuitry for signal processing, energy utilization, and system control.
- **Alert Mechanism:** Implemented automatic alerts for impact detection and emergency situations to improve workplace safety.
- **Practical Solution:** Focused on a low-cost, energy-efficient, and industry-oriented wearable safety device.

EXTRACURRICULAR & VOLUNTEER EXPERIENCE

IEEE UET Lahore Student Branch

Managing Director

Oct 2023 – April 2024

- Served as Managing Director at IEEE, leading teams, coordinating projects, and enhancing leadership, communication, and collaboration skills.

Blood Donor Society UET

Mentor

Aug 2023 – May 2024

- Worked as a mentor at BDS, guiding and managing an organization of 500+ members, coordinating events, and enhancing leadership, communication, and organizational abilities.

ADDITIONAL

Technical Skills: Microsoft Office (PowerPoint, Word, Excel), MATLAB, SIMULINK, HFSS, AutoCAD (Electrical), ETAP, Multisim, Proteus, VS Studio Code, Google Colab, Machine Learning, Deep Learning, Artificial Intelligence, Dev C++, Python, PyCharm

Non-Technical Skills: Leadership, Time Management, Team Collaboration, Problem Solving, Effective Communication, Decision Making

Interests: Innovative projects, Team Leadership, Research Work, On Site Work, Sports, Literature